PROJECT 02

HIMALAYAN SEISMIC BELT



SeismoWatch

OBSERVE THE EARTH. UNDERSTAND THE SIGNALS.

Real-time Earthquake Monitoring

Live data from official USGS Earthquake API

Focused on India & Himalayan Belt

Latitude 6°N – 38°N
Longitude 68°E – 98°E

Statistical Insights & Anomaly Detection

Understand patterns beyond individual events

02 The Problem

Earthquake data exists, but understanding the bigger picture is **not that simple.**

Distributed Data Sources

Earthquake information is available across multiple platforms like USGS, regional agencies and global seismic networks, often in different formats and with varying coverage.

Individual Records ≠ Bigger Patterns

A single earthquake record tells us the 'what', but not the 'why' or 'how unusual' it is. It's hard to see broader trends just by looking at individual events.

Need for Historical Analysis

Understanding unusual seismic activity requires historical data, statistical analysis, and pattern recognition over time, not just real-time events.





We need a unified, interactive, and analytical view of earthquake data

to move beyond individual events and understand the real patterns of seismic activity in the Indian Subcontinent and Himalayan region.

03 The Proposed Solution

SeismoWatch brings real-time earthquake data, historical analysis, and statistical anomaly detection into a single, interactive platform.

01 FETCH

Real Earthquake Data from USGS

Fetches real-time and historical earthquake observations from the official USGS Earthquake Catalog API.



02 STORE

Persistent & Structured Database

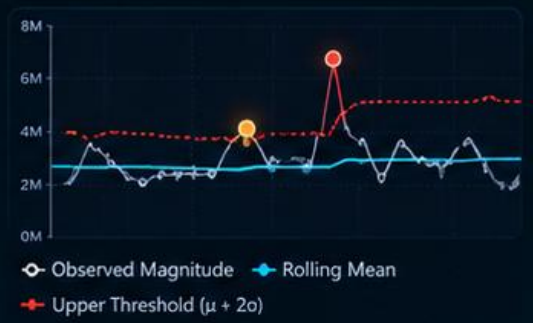
Validates, cleans, and stores earthquake records in PostgreSQL using Drizzle ORM with deduplication and ingestion tracking.



03 ANALYZE

Statistical Anomaly Detection

Applies prospective rolling-window analysis, calculates z-scores, and identifies magnitude outliers and frequency surges.



04 VISUALIZE

Interactive Insights & Exploration

Presents data through interactive maps, real-time dashboards, charts, and an anomaly laboratory for deeper exploration.



Real USGS Data

Live and historical observations from the official source.

Statistical Analysis

Magnitude outliers and frequency surge detection.

Focus on India & Himalayas

6°N – 38°N, 68°E – 98°E seismic belt.

All in One Platform

From raw data to insights, in a clean and interactive interface.

HIMALAYAN BELT
6°N – 38°N
68°E – 98°E

04 Technology Behind the Project

A modern, full-stack architecture to bring real earthquake data to an interactive and analytical platform.

Next.js + TypeScript

Modern, scalable web application with App Router.

Next.js 14 | TypeScript

USGS Earthquake API

Real-time and historical earthquake observations from the official USGS Earthquake Catalog.

GeoJSON | Global Data

PostgreSQL + Drizzle ORM

Persistent and structured database for earthquake records with type-safe ORM and migrations.

PostgreSQL | Drizzle ORM

Recharts + Leaflet

Interactive charts and geospatial maps for data visualization and exploration.

Recharts | Leaflet

Rolling-Window Z-Score Analysis

Identifies magnitude outliers and frequency surges using prospective statistical analysis.

Statistical Engine | Anomaly Detection

HOW IT ALL COMES TOGETHER

1. FETCH
Get real earthquake data from USGS API

2. STORE
Validate and save data in PostgreSQL using Drizzle ORM

3. ANALYZE
Apply rolling-window z-score analysis and frequency detection

4. VISUALIZE
Present insights through interactive maps, charts and dashboards



Core Analytical Engine

Prospective rolling mean and standard deviation with configurable window size and z-score threshold to detect statistically unusual seismic activity in the Indian Subcontinent and Himalayan region.

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6°N – 38°N, 68°E – 98°E

05

What Makes It Different?

More than just earthquake records — SeismoWatch provides deeper insights through statistical analysis, historical trends, and interactive visualization.

01 Prospective Rolling Baseline

Uses a rolling historical baseline (excluding the current event) to calculate z-scores and detect statistically unusual seismic activity.



02 Magnitude Anomaly Detection

Identifies magnitude outliers that significantly deviate from the expected historical pattern using statistical z-score analysis.

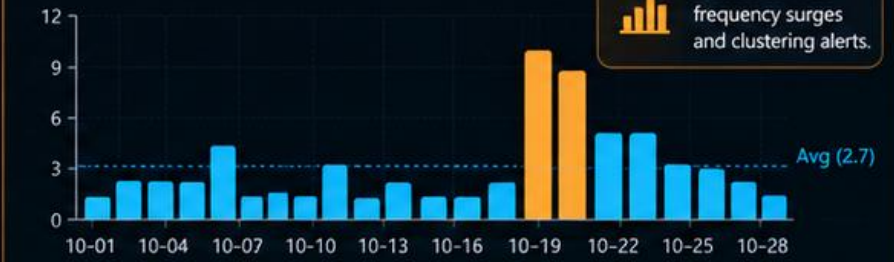
Detects unusually high or low magnitude events compared to regional historical behavior.

Z-Score Threshold | Magnitude Outliers | Statistical Significance

03 Independent Frequency Surge Detection

Identifies unusual clustering or swarms of earthquakes independently of magnitude outliers.

Detects 24-hour frequency surges and clustering alerts.



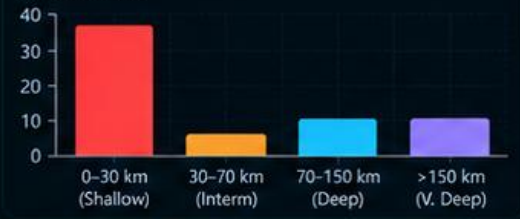
04 Historical Earthquake Exploration

Explore magnitude and depth distributions, daily trends, and time-series patterns over configurable time windows.

Magnitude Distribution (Gutenberg-Richter Trend)



Focal Depth Stratification (Hypocenter)



05 Interactive Geographic Visualization

View real earthquake locations on an interactive map with magnitude-scaled markers, hover details, and regional focus.



- Magnitude-scaled epicenter markers
- Interactive tooltips and event details
- Focused on India & Himalayan seismic belt
- Light & dark map layers (CartoDB via Leaflet)

06 Project Showcase

A live demonstration of SeismoWatch and its key features.



SeismoWatch

OBSERVE THE EARTH. UNDERSTAND THE SIGNALS.

Live Website
seismowatch-sigma.vercel.app

GitHub Repository
github.com/dk-khandelwal86/seismowatch

Connect on LinkedIn
linkedin.com/in/daksh-khandelwal-b02748391/

NEXT UP

Live Demonstration

Let's explore the actual website and see how SeismoWatch works!

